



API Specification Document

Telco Services

v1.0.0

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DEVELOPER API GUIDE DOCUMENT

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Revision History

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1.0 Introduction

This API specification document details the integration of Telco services that were previously managed by the Broker service in the Democratic Republic of Congo (DRC). The integration will now be executed through the **Transaction Processing Gateway**, commonly referred to as **GAG**. This strategic shift ensures a more robust and scalable solution for handling Telco transactions.

The Telco providers involved in this integration are:

- **Vodacom**
- **Africell**
- **Airtel**
- **Orange**

The services covered under this integration include:

- **Bank to Wallet Transaction (B2C):** Seamlessly transferring funds from a bank account to a mobile wallet.
- **Airtime Top-up:** Enabling the purchase of mobile airtime using bank funds.
- **Float Purchase:** Facilitating merchants in purchasing electronic float for distribution.

All request and response exchanges will be conducted in **XML format**, ensuring a consistent, structured, and reliable method of communication between systems.

For secure transactions, **signature verification** will be employed as the authentication mechanism, ensuring both the integrity and authenticity of all interactions.

Additionally, each calling channel will be required to provide a valid **certificate file (.cer)** for authentication purposes.

2.0 Bank to Wallet Service (B2C)

Method	POST
URL	http://10.11.6.110:8580/api/v1/banktowallet
Headers	content-type: application/xml
Authentication	signature

2.0.1 Request Field Description

No	FIELD	DESCRIPTION	MANDATORY	TYPE
1	rrn	Retrieval reference number. Unique Alphanumeric. Must be size 12	Y	String
2	amount	The transaction amount to be processed Minimum value > 0 Maximum decimal : 2	Y	Decimal
3	bankId	The instance bank code. Will default DRC bankId, 43 if field is null	N	String
4	currency	Transaction currency. Will default to USD if field is null	N	String
5	sender	Sender mobile number/ account	Y	String
6	name	Sender full name	Y	String
7	number	Receiver mobile number / account	Y	String
8	debitAccount	Source account number	N	String
9	telco	Telco name. Valid values: 1. ORANGE-MONEY 2. AIRTEL-MONEY 3. VODACOM-MONEY 4. AFRICELL-MONEY	Y	String
10	channel	Calling channel Id. Defaults to OMNI if field is null	N	String

11	tranID	Wallet transaction ID	N	String
12	chargeAmount	Charges / Fees	N	String
13	chargeFlag	Mark transaction with charges. Valid values: a. Y b. N	N	String
14	callbackUrl	Channel call back url	N	String
15	optIn	Indicator if the sender has opted into sms notification services (0 = No, 1 = Yes)	N	String
16	destinationShort Code	Destination short code	N	String
17	signature	Encrypted signature for security and authentication See Digital Signature Section	Y	String

2.0.2 Response Field Description

NO	FIELD	DESCRIPTION	TYPE
1	msgId	Request RRN value	String
2	status	Transaction status. a. Sample success status: i. OK 1234 , where 1234 is the transaction ID b. Sample failure status i. ERROR 409 , where 409 is the error status code	String
3	responseCode	Status response code	String
4	responseDesc	Status response description	String

2.0.3 Sample Request

```
<?xml version="1.0" encoding="utf-8"?>
```

```

<request                                xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <eazzycash>
    <rrn>123456789101</rrn>
    <amount>100.00</amount>
    <bankId>43</bankId>
    <currency>USD</currency>
    <sender>0720123124</sender>
    <name>MICHAEL ANDAMA</name>
    <number>999967426</number>
    <debitAccount>1234</debitAccount>
    <telco>AIRTEL_MONEY</telco>
    <channel>OMNI</channel>
    <tranID>1234</tranID>
    <chargeAmount>5.00</chargeAmount>
    <destinationShortCode>1234</destinationShortCode>
    <chargeFlag>Y</chargeFlag>
    <callbackUrl>http://localhost:8080</callbackUrl>

    <signature>jZzbRrd8Ycwg8xClrOdddKZcHjXHnUtgf6FA9gMFtCMF720S8rZPiVgtc0nPBuN
Xa4WIYvK+elzapEDbxUvA+oiyWtPZYM+fql/UAXmJcTjJivr8z7ztjOx0Tq7La63Yvju06g6AWjhs3LrZ
a8ECBqQkQNdk9DBTcBT/oSP0Nla3IHq/WREA40ttMPtgbQoCmp/ZCL7Rmw8ulm/A2VEUvM14L
Efn2jMzij1DWPzidsCkA6q7CYPPHLFIBVpevEpTNp9WTm80RWYr8oZx5jT8i4FAFMnN58kXd1/Q
HfjgMxvfkP+T3XmFg2Kv5PdF7rOF0XvsaY9u6hfy2NbnQwaSMg==</signature>
    <optIn>0</optIn>
  </eazzycash>
</request>

```

2.0.4 Sample Success Response

```

<response>
  <msgId>202409190124</msgId>
  <eazzycashres>
    <status>OK|100100</status>
    <responseCode>000</responseCode>
    <responseDesc>Success.</responseDesc>
  </eazzycashres>
</response>

```

2.0.5 Sample Failure Response

```

<response>
  <msgId>202409190124</msgId>

```



```

<eazzycashres>
  <status>ERROR|409</status>
  <responseCode>409</responseCode>
  <responseDesc>de-dup! the request is a duplicate and has already been
processed.</responseDesc>
</eazzycashres>
</response>

```

3.0 Airtime Top-up Service

Method	POST
URL	http://10.11.6.110:8580/api/v1/airtimetopup
Headers	content-type: application/xml
Authentication	signature

3.0.1 Request Field Description

No	FIELD	DESCRIPTION	MANDATORY	TYPE
1	rrn	Retrieval reference number. Unique Alphanumeric. Must be size 12	Y	String
2	amount	The transaction amount to be processed Minimum value > 0 Maximum decimal : 2	Y	Decimal
3	country	The instance bank code. Sample code: a. CD for DRC	Y	String
4	currency	Transaction currency. Will default to USD if field is null	N	String
5	msisdnA	Receiver mobile number	Y	String
6	telcom	Telco name. Valid values: 1. ORANGE 2. AIRTEL 3. VODACOM 4. AFRICELL	Y	String

7	channel	Calling channel Id.	Y	String
8	signature	Encrypted signature for security and authentication See Digital Signature Section	Y	String

2.0.2 Response Field Description

NO	FIELD	DESCRIPTION	TYPE
1	return	Transaction status. c. Sample success status: i. OK 1234 , where 1234 is the transaction ID d. Sample failure status i. ERROR 409 , where 409 is the error status code	String

3.0.3 Sample Request

```
<?xml version="1.0" encoding="UTF-8"?>
<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/">
  <s:Body xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns:xsd="http://www.w3.org/2001/XMLSchema">
    <topup xmlns="http://business.airtime.telcos.equitybank.com/" xsi:type="topup">
      <rrn xmlns="">123456789012</rrn>
      <msisdnA xmlns="">2547200000000</msisdnA>
      <amount xmlns="">100.0</amount>
      <currency xmlns="">USD</currency>
      <telcom xmlns="">Orange</telcom>
      <country xmlns="">KE</country>
      <channel xmlns="">Omni</channel>
      <signature
        xmlns="">aaBUjKr0V7N7Mnzc3EicpQq14XPWF/Dns4l5nFz05DvGnpZss9JFXngrQ0nDLxKnn2y6
        L8bEhSH+TGy4bIKCRKYRnNjJfFOGuzDTRCChjmaBEzcJMVSWK5+8F6YfEPJqGoM08Op73Kj2
        cOVHVoYAqit+DKojJRijbKtxZW1ZpyZC2F60Cjqf1prJglSRhEdjnXGpzVto+k2siGSXUV35mYCL8
        Hw0OOTV1O/Qz1HH3/YzR9oRXejzVaVGghx/RqCFMy2UdUvPfEdyVyZzYxsg2kp+Zdf8ufj1GTwT
        oUQu207AreyPCye0LUN+jtQiD6+jHy2ENFpcfnulbv41hWz3A==</signature>
      </topup>
    </s:Body>
  </s:Envelope>
```

3.0.4 Sample Success Response

```
<Envelope>
  <Body>
    <topupResponse>
      <return>OK|100100</return>
    </topupResponse>
  </Body>
</Envelope>
```

3.0.5 Sample Failure Response

```
<Envelope>
  <Body>
    <topupResponse>
      <return>ERROR|500</return>
    </topupResponse>
  </Body>
</Envelope>
```

4.0 Float Purchase Service

Method	POST
URL	http://10.11.6.110:8580/api/v1/cashallocation
Headers	content-type: application/xml
Authentication	signature

4.0.1 Request Field Description

No	FIELD	DESCRIPTION	MANDATORY	TYPE
1	rrn	Retrieval reference number. Unique Alphanumeric. Must be size 12	Y	String

2	amount	The transaction amount to be processed Minimum value > 0 Maximum decimal : 2	Y	Decimal
3	bankId	The instance bank code. Sample value: a. 43 for DRC b. 54 for Kenya	Y	String
4	sender	Sender mobile number/ account	Y	String
5	name	Sender full name	Y	String
6	number	Receiver mobile number / account	Y	String
7	telco	Telco name. Valid values: 1. ORANGE 2. AIRTEL 3. VODACOM 4. AFRICELL	Y	String
8	destinationShort Code	Destination short code	Y	String
9	storeNumber	Store number	Y	String
10	channel	Calling channel Id. Sample values: a. OMNI b. STK	Y	String
11	tranType	Transacation type. Sample value a. BUYFLOAT	N	String
12	refNumber	Same as rrn	N	String
13	msgId	Same as rrn	Y	String
14	signature	Encrypted signature for security and authentication See Digital Signature Section	Y	String

4.0.2 Response Field Description

NO	FIELD	DESCRIPTION	TYPE
1	msgld	Request RRN value	String
2	status	Transaction status. e. Sample success status: i. OK 1234 , where 1234 is the transaction ID f. Sample failure status i. ERROR 409 , where 409 is the error status code	String
3	responseCode	Status response code	String
4	telcoResponseDesc	Status response description	String
5	telcoResponseCode	Response code from telco	String
6	tranID	Transaction Id for when status is Success / OK	String

4.0.3 Sample Request

```
<?xml version="1.0" encoding="UTF-8"?>
<request>
  <msgld>20240919876</msgld>
  <eazzycash>
    <bankld>43</bankld>
    <amount>1000</amount>
    <sender>254798264668</sender>
    <name>WILLY KIPROTICH TONGEN</name>
    <number>999119410</number>
    <telco>AIRTEL</telco>
    <destinationShortCode>45554</destinationShortCode>
    <rrn>20240919920</rrn>
    <storeNumber>4444555</storeNumber>

    <signature>GeZ3KLe4rBhDqXiDSV8z8Alc1tqgJ0B62B8XpFmG2pXxCFzzDGhX6f4uibTtvdXnOJa
TGf6Qh4Wl6uSG5ogrX7aszG/VgZvk0pyF8zloDT+OBI5RIsbUUTyDy/UZXV8ECaI3+3g7Qzx
vE9euSm7tW69UlifBlSZktd8E3yFOtvBTcEHKe4XsjgsZyBZfGlgY/e1n89F7e7hHq6VtRUBcLylqUM
kjzd1q87jRnJ6oVhooUR2Jc92V/GDdZQ7LTuaVSzV15YaG9mAKPR3rROokMyxT750eV5aN1
tgByzBqlsFukXMUsoUpxhe+Z0/tzxP824hV6YSjJI0JVZZhwgakovA==</signature>
```

```
<channel>OMNI</channel>
<tranType>BUYFLOAT</tranType>
<refNumber>254798264666</refNumber>
<source>OMNI</source>
</eazzycash>
</request>
```

4.0.4 Sample Success Response

```
<response>
  <msgId>1234444</msgId>
  <eazzycashres>
    <status>OK</status>
    <responseCode>0</responseCode>
    <tranID>12333444</tranID>
    <telcoResponseCode>0000000</telcoResponseCode>
  </eazzycashres>
</response>
```

4.0.5 Sample Failure Response

```
<response>
  <msgId>202409241042</msgId>
  <eazzycashres>
    <status>ERROR</status>
    <responseCode>-1</responseCode>
    <telcoResponseCode>DP02050001002</telcoResponseCode>
    <telcoResponseDesc>Dear customer,the transaction amount you have entered is insufficient.
Please call 1211 for help, thank you.</telcoResponseDesc>
  </eazzycashres>
</response>
```

5.0 Digital Signature Generation

To ensure secure communication, the calling channel must provide a valid certificate file(**.cer**) certificate file.

The digital signature is generated using the **SHA256withRSA** algorithm and passed as a **Base64-encoded** value in the signature field of the API request.

5.0.1 Signature Generation

1. **Provide a Valid Certificate:**
 - The calling channel must supply a **.cer** certificate file.
2. **Generate the Digital Signature:**
 - **Concatenate the clear text fields** (detailed below).
 - **Format the amount** as a **BigDecimal** with two decimal places using `setScale(2, RoundingMode.CEILING)`.
 - **Hash** the concatenated string using **SHA-256**.
 - **Encrypt** the hash with **RSA** using the private key.
 - **Encode** the encrypted hash as a Base64 string.
 - **Pass** the Base64 signature in the signature field.

5.0.1 Clear Text Values

For each transaction type, the clear text values must be concatenated in the specified order. The amount field must be a **BigDecimal** with two decimal places.

1. **Bank to Wallet (B2C) Transaction:**
 - Concatenate the following fields in this order:
amount + sender + name + number + telco + rrn
 - Ensure the amount is a **BigDecimal** and formatted as `setScale(2, RoundingMode.CEILING)`.
2. **Airtime Purchase Transaction:**
 - Concatenate the following fields in this order:
phoneNumber + amount + telco + channel + reference
 - Ensure the amount is a **BigDecimal** and formatted as `setScale(2, RoundingMode.CEILING)`.
3. **Float Purchase Transaction:**
 - Concatenate the following fields in this order:
amount + sender + name + number + telco + rrn
 - Ensure the amount is a **BigDecimal** and formatted as `setScale(2, RoundingMode.CEILING)`.